STA2020 Tutorial 2: Multiple Linear Regression

Box and Youle (1955) performed a series of 19 experiments to investigate a chemical reaction. The aim was to establish if a relationship exists between the percentage of material converted to the desired substance (y) and the temperature of the reaction (in °C), the time of the reaction (in minutes), and the concentration of the element in question (in %). The results are shown in the following table:

Experiment	%Converted (Y)	Temperature (X_1)	Concentration (X_2)	Time (X_3)
1	45.9	162	23	3
2	53.3	162	23	8
3	57.5	162	30	5
4	58.8	162	30	8
5	60.6	172	25	5
6	58	172	25	8
7	58.6	172	30	5
8	52.4	172	30	8
9	56.9	167	27.5	6.5
10	55.4	177	27.5	6.5
11	46.9	157	27.5	6.5
12	57.3	167	32.5	6.5
13	55	167	22.5	6.5
14	58.9	167	27.5	9.5
15	50.3	167	27.5	3.5
16	61.1	177	20	6.5
17	62.9	177	20	6.5
18	60	160	34	7.5
19	60.6	160	34	7.5

The correlations between all the variables are as follows:

	Converted	Temperature	Concentration	Time
Converted	1.000	0.404	0.080	0.393
Temperature	0.404	1.000	-0.462	-0.022
Concentration	0.080	-0.462	1.000	0.177
Time	0.393	-0.022	0.177	1.000

A multiple linear regression was performed on this dataset, yielding the following results:

```
Call:
lm(formula = Converted ~ Temperature + Concentration + Time,
    data = MLR)

Residuals:
    Min      1Q  Median      3Q     Max
-8.2075 -2.0557  0.1633  3.2712  4.7503

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -26.0353    32.9734  -0.790   0.4421
Temperature     0.4046     0.1747   2.316   0.0351 *
Concentration   0.2930     0.2604   1.125   0.2783
Time            1.0338     0.5994   1.725   0.1051
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.078 on 15 degrees of freedom
Multiple R-squared:    A   ,     Adjusted R-squared:    B
F-statistic: 3.027 on 3 and 15 DF,  p-value: 0.06235
```

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	151.002	50.334	3.027	0.062
Residual	15	249.462	16.631		
Total	18	400.464			

Use the above provided information to answer the following questions:

1) Give the estimated regression equation.
2) According to this model, what will the predicted percentage of converted material be in an experiment where a concentration of 28% is run for 6 minutes at 170°C?

3) According to the model, how much longer does the experiment need to be run in order to increase the percentage of converted material by 25%? Evaluate the feasibility of this answer.
4) Is there a sensible interpretation of the estimate for the intercept coefficient?
5) Calculate the missing values in the R output (specifically the value of A and B). Additionally, based on this, provide an estimate of the sample correlation coefficient.
6) Comment on the overall significance of this fitted regression model by referring to the appropriate values.
7) Which of the independent variables in the fitted model are significant at α = 0.05?
8) Is there any danger of multicollinearity in this analysis? Justify your answer by referring to the relevant values.
9) How would you go about trying to improve the model fit?

As an extra exercise, use the R script (containing the dataset) and perform another multiple regression analysis – but excluding the least significant variable. Compare this new model to the results given above.